Is there a cheaper corner than (degree 5, 4 colors)?

12 of 21 (degree, n_colors) cells x 6 sizes, 72 removal rows and 32 random rows

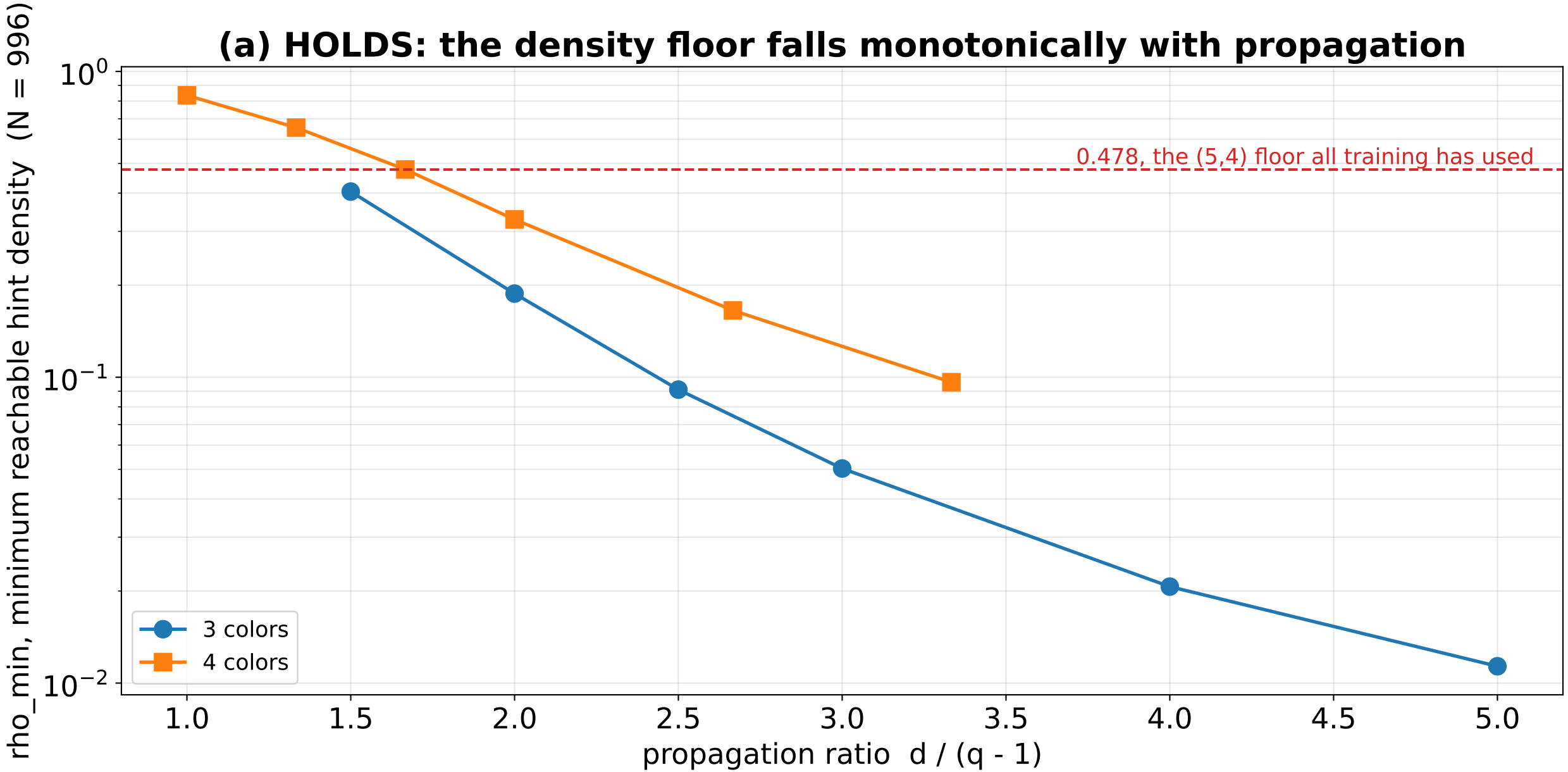
sudoku_dd_control_param_2026-10-24

YES, and at N = 96 it is worth about a factor of 8 in N.

(d=8, q=4) reaches mean deduction depth 19.1 near its density floor at N = 96, against 9.3 for (d=5, q=4) at the same size. The baseline does not reach 19.1 until N ~ 813.

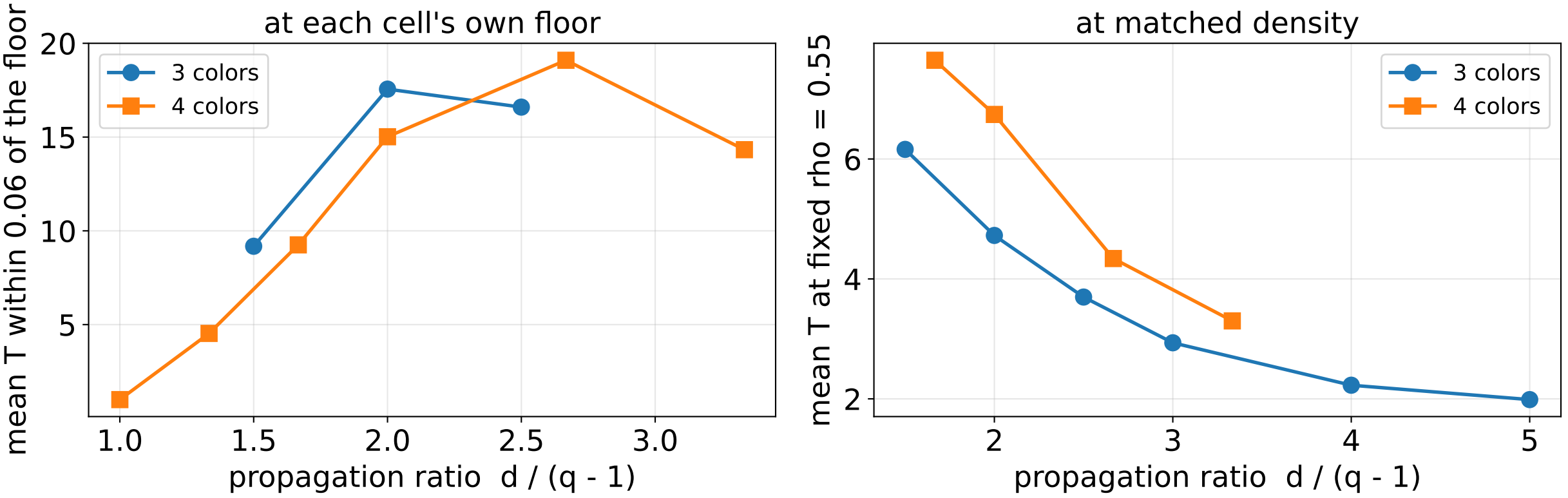
But NOT for the reason predicted, and rho_min is still not a critical point: T grows as a power of N at every density DD can act on.

The propagation ratio d/(q-1) governs how many neighbours there are per colour that has to be excluded before DD can force a node. Prediction (a) was that rho_min falls as that ratio rises; prediction (b) was that the DEEPEST instances live at low ratio. The first holds across three decades; the second is backwards.



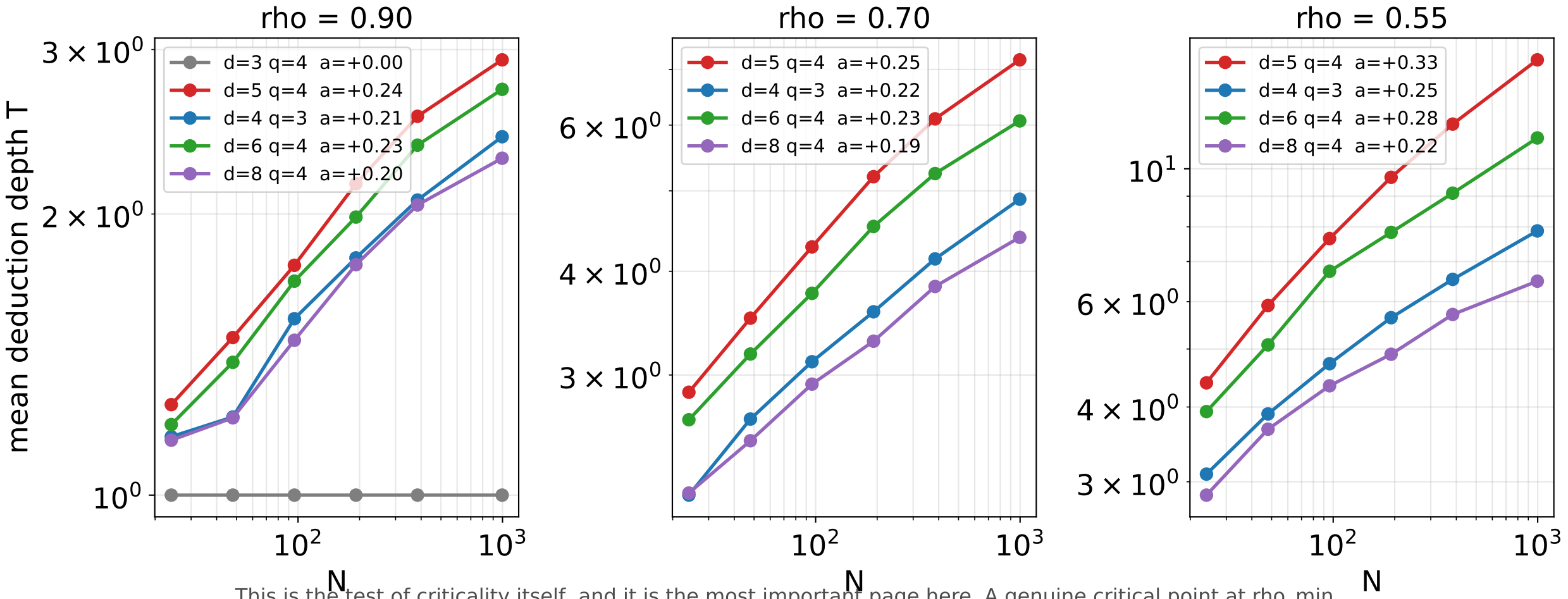
Measured at the largest size, where rho_min is already asymptotic. It spans nearly two decades, 0.84 at ratio 1.00 down to 0.011 at ratio 5.00, and falls monotonically throughout -- so prediction (c)'s falsifier, that rho_min is flat in (d,q) and there is no cheap corner, is itself refuted. Ratio 1.00 is the degenerate case d = q-1, where DD can force almost nothing and greedy removal stalls at 84% hinted.

(b) FAILS, backwards: depth peaks at HIGH propagation, not low



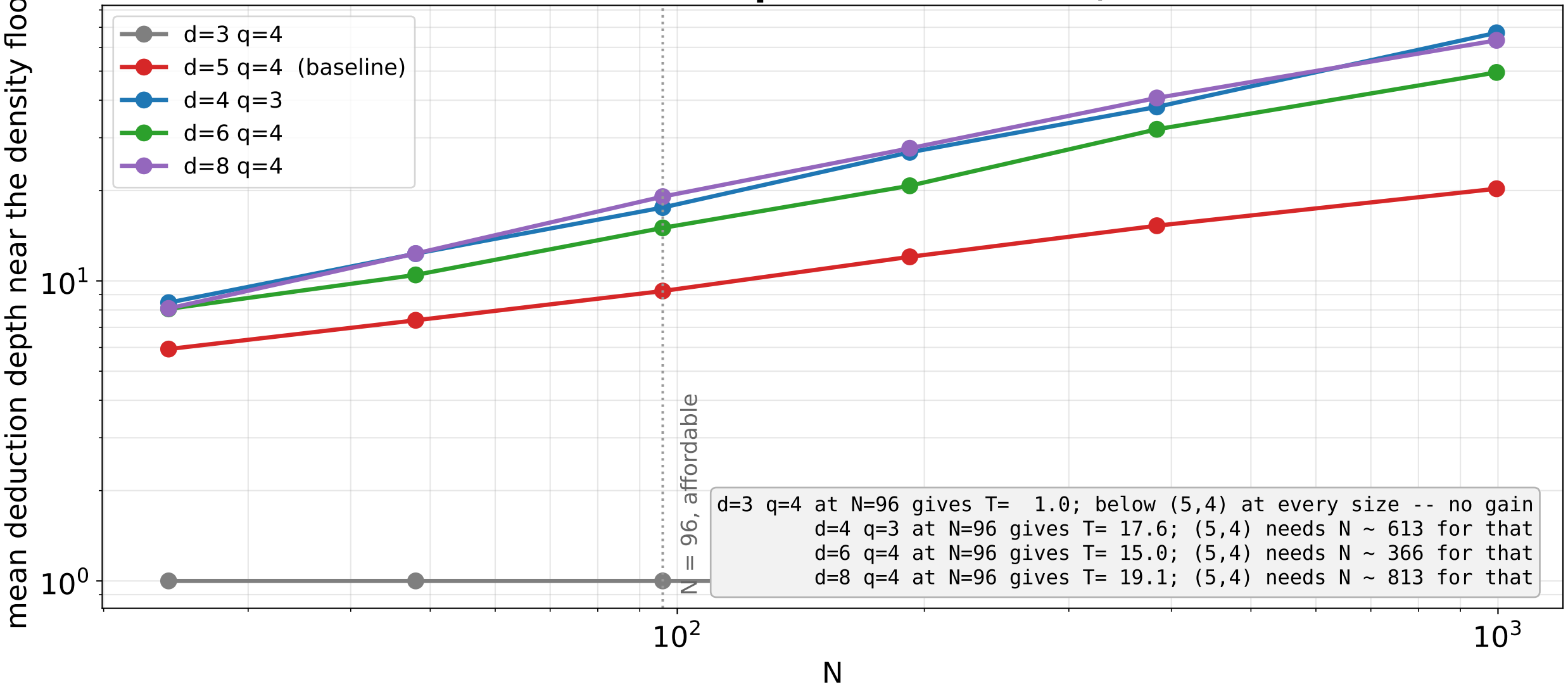
(b) expected a ratio <= 1.2 cell to reach mean T >= 22 at N=96. Those cells are d = q-1, the SHALLOWEST on the grid: DD forces nothing there, removal stalls at once, T ~ 1. Depth has an interior optimum instead -- low ratio stalls removal early, high ratio forces so many nodes per round that few rounds are needed, and the longest chains sit between. The panels disagree on where, and that is the caveat on the whole sweep: at each cell's OWN floor the high-ratio cells win, but at MATCHED density the baseline (5,4) is the deepest thing on the grid. The cheap corner buys depth by moving the floor down, not by making deduction harder.

(d) FAILS: T grows as a power of N at every density DD can act on

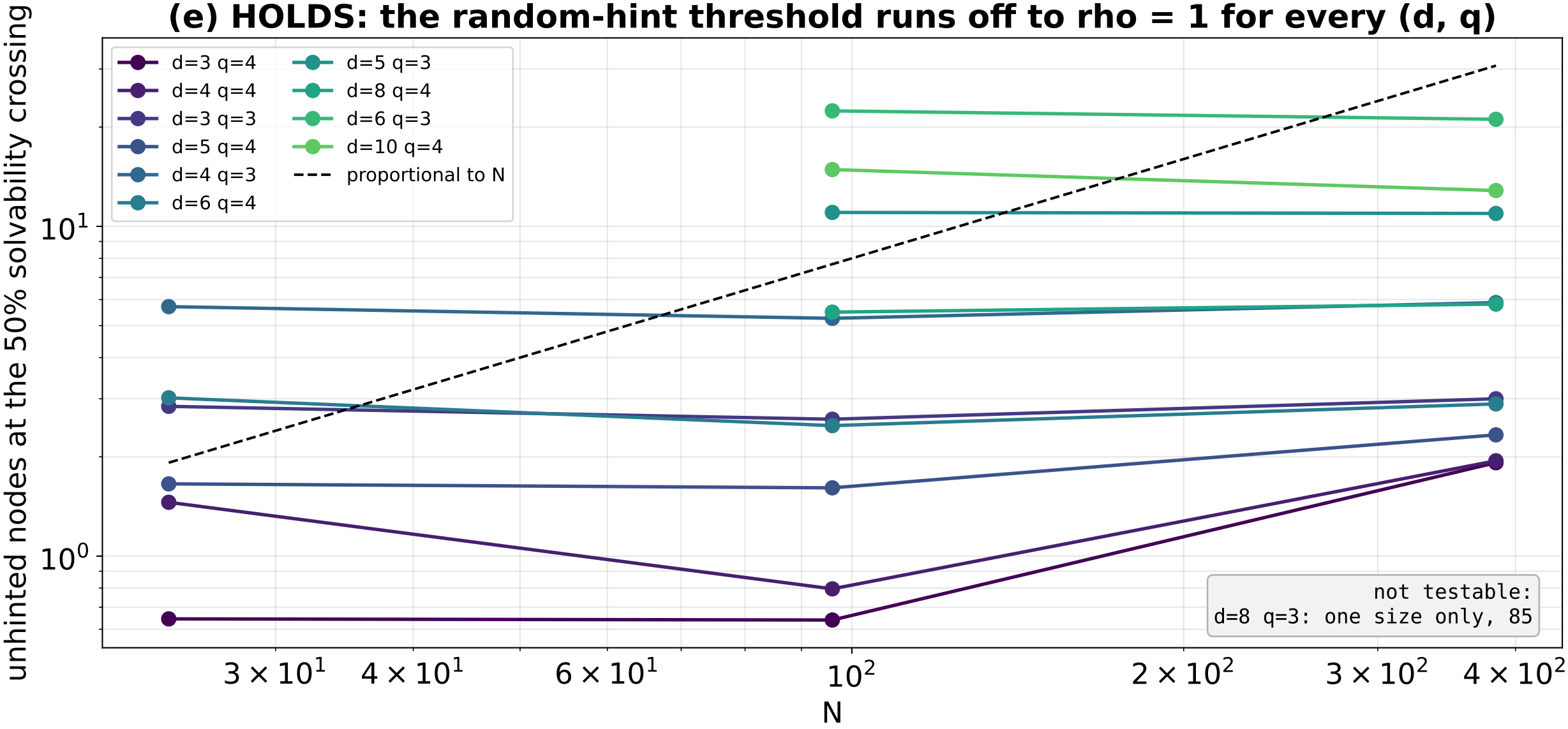


This is the test of criticality itself, and it is the most important page here. A genuine critical point at rho_min requires T to grow with N near the floor while staying N-INDEPENDENT far from it. Instead the exponent a in T ~ N^a is positive at rho = 0.90 in every cell that deduces anything -- it never reaches zero, it only gets smaller. The one exception, d=3 q=4 at a = +0.00, is not evidence for criticality but the opposite: it is pinned at exactly T = 1.0 at every N because d = q-1 lets DD force nothing beyond the first round, so there is no chain to lengthen. Excluding it, rho_min is where greedy removal stops rather than a critical point, across the whole (degree, n_colors) grid and not at (5,4) alone -- a real strengthening of finding 1 rather than a restatement of it.

What the cheap corner is worth, in N



Read with the caveat from the depth-vs-ratio page: these cells reach their depth at their OWN floor, which for (8,4) is rho ~ 0.17 against 0.48 for the baseline, so the instances are far sparser. Whether that is a fair substitute for large N is a question about what the transformer is being asked to learn, not one this sweep answers. Token cost favours it twice over: q=3 needs a smaller vocabulary and d=4 a shorter graph section than the baseline's d=5.



If the random-hint ensemble had a finite-density solvability transition, the number of unhinted nodes at the 50% crossing would grow proportionally to N, like the dashed line. It is flat in N in every cell with two or more measurable sizes. The CONSTANT rises steeply with degree -- about 3 at d=3 and 85 at d=8, q=3 -- which is the propagation ratio again, not a transition; the earlier reading of that 85 as pressure on this prediction confused a large constant for N-growth. The cells with one size are nan at the others because the crossing fell below the rho = 0.50 grid floor, not because it is absent: extending the grid downward for ratio >= 4 is the fix, and it is cheap.

Scorecard against the pre-registered predictions

(a)	**HOLDS**	rho_min falls monotonically with d/(q-1), 0.84 -> 0.011 over ratio 1.00 -> 5.00. The specific claim rho_min(3,4) > 0.477 was met at 0.836; the d=10 q=4 half awaits the rest of the grid.
(b)	**FAILS**	Backwards. The ratio <= 1.2 cells are d = q-1, the SHALLOWEST on the grid (T ~ 1, floor at 84% hinted), not the deepest. Depth has an interior optimum in the ratio.
(c)	**REFUTED**	Its falsifier required rho_min and max T flat within +-20%. They span two decades and a factor of 100. There IS a cheap corner -- which is the useful half of this sweep.
(d)	**FAILS**	The exponent in T ~ N^a is positive at rho = 0.90 in every cell that deduces at all (+0.19 to +0.25), so nothing is critical at rho_min anywhere on the grid. The sole flat cell, d=3 q=4, is pinned at T = 1.0 for lack of any chain to grow. Generalises finding 1 from (5,4) to all of (degree, n_colors).
(e)	**HOLDS**	unhinted-at-crossing is flat in N in every cell with two or more sizes. The large constants at high degree are the ratio, not a transition.

12 of 21 cells and 72 of 126 removal rows are in. (a), (c) and (d) are decided by what is here and the rest of the grid cannot overturn them. (b) could still be rescued by a ratio <= 1.2 cell at higher q, which is what remains -- treat it as very likely failed rather than settled.